TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
Substitution: get a variable alone, put its expression into the other equation, solve, substitute back.
Variable drops out? False statement (like 0 = 7): no solution. True statement (like 5 = 5): infinite.
Check: put both values into both equations.

1. Solve: $y = 2x + 1$ and $4x - 2y = -2$.

Tip: If the variable drops out, read the statement.

A. No solution

B. (0, 1) only

C. Infinitely many solutions

2. Floor seats cost \$10 more than balcony seats. 300 floor seats and 200 balcony seats bring in \$8000. Which system models this?

Tip: Which price is bigger? Match each count with its price.

F. $f = b + 10$ and $300f + 200b = 8000$

G. $b = f + 10$ and $300f + 200b = 8000$

H. $f = b + 10$ and $200f + 300b = 8000$

3. Solve the concert system from Question 2. What does each seat cost?

> **Tip:** Substitute b + 10 for f.

300(_____) + 200b = 8000

500b + 3000 = 8000, so b = _____

f = b + 10 = _____